A hierarchy of propositions: deciding which level to address in casework

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In a previous paper in this journal (Cook R, Evett IW, Jackson G, Jones PJ and Lambert JA. A model for case assessment and interpretation. Science & Justice 1998; 38: 151–156), the authors described a model for case assessment and interpretation. An essential element of the model, indeed of any interpretation, is that of framing a pair of propositions to weigh against each other. It is one of the functions of the scientist to determine, in the light of the evidence that is found and of the circumstances of the case as they appear at the time, the pair of propositions that can realistically be addressed. It is helpful to consider a broadly-defined "hierarchy of propositions". In general, the higher the level of the propositions, the greater the assistance that will be given to the court but, of course, the scientist may not stray outside the bounds of his/her expertise nor, indeed, into the realms of advocacy. This paper, a continuation of the first, discusses the hierarchy of propositions by means of case examples.

Dans un article précédent publié ici (Cook R, Evett IW, Jackson G, Jones PJ et Lambert JA. A model for case assessment and interpretation. Science & Justice 1998; 38: 151–156), les auteurs décrivent un modèle pour une évaluation de cas et l'interprétation. Un élément essentiel du modèle, et de toute interprétation, est de cibler deux propositions qu'il faut soupeser l'une par rapport à l'autre. C'est l'une des fonctions du scientifique que de déterminer les deux propositions réalistes à considérer, à la lumière des indices trouvés et des circonstances du cas telles quelles apparaissent à ce moment-là. Il est utile de considérer "une hiérarchie de propositions" grossièrement définies. En général, plus le niveau des propositions est élevé plus l'aide au tribunal est grande mais, évidemment, le scientifique ne doit pas dévier hors de son champ d'expertise, ni dans le domaine du plaidoyer. Cet article, qui est une suite directe du précédent, discute la hiérarchie de propositions au moyen d'exemples de cas.

Die Autoren haben in einer vorangegangenen Arbeit (Cook R, Evett IW, Jackson G, Jones PJ and Lambert JA. A model for case assessment and interpretation. Science & Justice 1998; 38: 151–156) ein Modell zu Abschätzungs- und Interpretationsfragen in der Fallarbeit beschrieben. Ein entscheidendes Element des Modells – und eigentlich jeder Interpretation – ist das Aufstellen und gegenseitige Abwägen von Hypothesen. Der Wissenschaftler hat dabei die Aufgabe jene Hypothesen zu bestimmen, die unter Berücksichtigung der Art des Beweismittels und der Umstände des Falles wirklich angewendet werden können. Dabei ist es hilfreich von einer eher weitgefaßten „Hierarchie von Hypothesen" auszugehen. Grundsätzlich gilt: je allgemeiner die Hypothesen gefaßt sind, um so größer ist ihre Hilfe für das Gericht. Natürlich dürfen sie weder außerhalb des Wissensgebietes des Sachverständigen liegen noch die Bereiche berühren, die Anwälten vorbehalten sind. Die Arbeit diskutiert in Fortsetzung der ersten, die Hierarchie von Hypothesen an Fallbeispielen.

En un trabajo anterior publicado en esta revista (Cook R, Evett IW, Jackson G, Jones PJ and Lambert JA. A model for case assessment and interpretation. Science & Justice 1998; 38: 151–156) los autores describían un modelo para la valoración e identificación de casos. Un elemento esencial del modelo, en realidad de toda interpretación, es el enmarcar dos propuestas para ser valoradas una frente a otra. Una de las funciones del científico es determinar, a la luz de la evidencia encontrada y de las circunstancias del caso tal como aparecen en ese momento el par de propuestas que pueden ser realisticamente consideradas. Es aconsejable considerar una 'jerarquía de propuestas' que esté ampliamente definida. En general, cuanto más alto sea el nivel de las propuestas, mayor será el apoyo prestado al tribunal aunque desde luego el científico no debe salirse de los límites de su pericia, ni meterse en el terreno de la abogacía. Este trabajo, continuación del primero, plantea la jerarquía de las propuestas a través de ejemplos de casos reales.

Key Words: Forensic science; Bayesian inference; Likelihood ratio; Probability; Statistics; Decision-making.

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Introduction

A previous paper [1] explained that, to interpret evidence, a forensic scientist should consider different propositions which represent, respectively, prosecution and defence allegations in a criminal trial. There has always been a tendency, in the adversary system of justice, for the scientist called by the prosecution to express an opinion with regard to the prosecution proposition and then to respond to alternative propositions put by the defence. There are times when this, probably unfairly, makes the scientist appear to be partisan – defending the prosecution position against attack. It is recognised that a scientist should consider defence propositions, but the culture has tended to create an atmosphere in which the scientist takes a position and then, in preparation for court, thinks of all of the likely propositions to be put forward by defence and mentally prepares responses to them. The training of new scientists generally includes questions from his/her seniors of the kind "and what would you say if defence allege so and so?"

The model which was discussed in the previous paper [1] requires that the framing of alternative propositions be addressed during the earliest stages of a case examination, but it is rarely a trivial exercise to do this. Indeed, in the training workshops that are held within the FSS on evidence interpretation it is the stage of framing alternatives to address that causes the most difficulty. Nevertheless, the effort is worthwhile because scientific interpretation of evidence should be logical and the establishment of clearly formulated alternative propositions is a crucial basis for a logical view. Furthermore, it places an emphasis on a balanced view of the evidence.

The essence of this work is related in many ways to the powerful methods for formalising legal reasoning laid down by Wigmore [2], Schum [3] and Kadane and Schum [4]. One of the aims of the project which led to this and the previous paper is to make such seminal ideas accessible to practising forensic scientists by the use of case based examples.

This paper is concerned with the framing of alternatives, with the idea of a "hierarchy of propositions" as central to the solution of the problems. The next section explains what is meant by the phrase and the ideas are then developed by means of examples.

The hierarchy of propositions

It is a fundamental principle that it is not possible for a scientist to speculate on the truth of a proposition without considering at least one alternative proposition. In the forensic sphere, this often conveniently reduces to addressing one proposition which represents the prosecution view and one which represents the defence view. In practice, the propositions that are addressed will depend on the circumstances of the case, the observations that have been made, background data that are available and the domain of expertise of the scientist. This means that the scientist must often settle for addressing propositions which are some way removed from those that the jury must address. It is helpful to regard pairs of propositions addressed in this way as falling into three broad generic classes, which are here called *Source, Activity* and *Offence* and designated with Roman numerals I to III, respectively. The levels are shown in Table 1 in reverse numerical order to emphasise that progress down the table implies movement down the hierarchy.

TABLE 1 Examples of the hierarchy of propositions.

Level	*Generic*		*Examples*
III	Offence	A	Mr A committed the burglary
			Another person committed the burglary
		B	Mr B raped Ms Y
			Some other man raped Ms Y
		C	Mr C assaulted Mr Z
			Mr C had nothing to do with the assault of Mr Z
II	Activity	A	Mr A is the man who smashed window X
			Mr A was not present when window X was smashed
		B	Mr B had sexual intercourse with Ms Y
			Some other man had sexual intercourse with Ms Y
		C	Mr C is the man who kicked Mr Z in the head
			Mr C was not present at the kicking of Mr Z
I	Source	A	The glass fragments came from window X
			They came from some other broken glass object
		B	The semen came from Mr B
			The semen came from some other man
		C	The blood on Mr C's clothing came from Mr Z
			The blood on Mr C's clothing came from an unknown person

The bottom line of the table contains the first, usually the simplest, level of propositions and these relate to the *source* of evidentiary material. The table shows three examples. The assessment of level I pairs of propositions such as these will be based on observations, measurements and analyses. The probability of the evidence given the first proposition of the pair will be considered from a careful comparison between two samples. The probability of the evidence given the second proposition of the pair will be considered by comparison, in principle at least, between one of the samples and some kind of population of alternative sources. In example IA the scientist might refer to data from a reference collection of glass objects. In each of the examples IB and IC, he will calculate a match probability based on allele proportions estimated from a reference database.

Moving up the table, level II is the class of propositions which relate to *activities*. Addressing these propositions also requires observations, measurement and analysis, but now there are further issues to take into account. To address the first of each pair, in addition to comparison of observations of any recovered material with control samples, it is necessary to consider the probability of evidential transfer and persistence. Questions arise of the kind: "is this the sort of staining I would see if Mr C is the man who kicked Mr Z?"; "what is the probability of finding this quantity of matching glass if Mr A is the man who smashed the window X?". To address the second proposition of each pair, questions arise of the kind: "is this the sort of staining I would see if Mr C was not present at the kicking of Mr Z?"; "what is the probability of finding this quantity of matching glass if Mr A was not present when window X was smashed?". Here, too, background surveys can be useful. In relation to the former question, for example, the survey of Briggs [5] of bloodstains on clothing might provide useful guidance. In relation to the latter, surveys of glass found on clothing such as those by McQuillan and Edgar [6] and Lambert, Satterthwaite and Harrison [7] are useful.

One distinction between level I and level II propositions is that interpretation with regard to pairs of the latter type do not necessarily depend on transfer material being found. For example, the corresponding level II examples could be addressed in the event of, respectively: no glass fragments on Mr A's jacket; no bloodstaining on Mr C's clothing.

Another, important difference is that level I propositions generally require little in the way of circumstantial information, but level II propositions cannot be addressed without a framework of circumstances. For example, the transfer/persistence issues relevant to the question "what is the probability of finding this quantity of matching glass if Mr A is truly the man who smashed the window X?" require information about how the window was smashed and the time interval between the incident and the taking of the clothing. Likewise, to address the question "what is the probability of finding this quantity of matching glass if Mr A were unconnected with the smashing of the window?" then it is necessary to know if there is something about Mr A which would predispose him to have glass on his clothing. In the absence of such information, assumptions will need to be made about the relevance of either the McQuillan/Edgar or Lambert et al surveys. The more information that is available to the scientist, then the more effective he/she can be in exercising judgement in relation to level II propositions. In this regard there needs to be a degree of interaction between scientist, investigator and/or advocate which is probably not required for level I propositions.

It is also a principle of this approach that, because the interpretation depends on the framework of circumstances as they appear to the scientist, it is necessary that they should be clearly itemised in the statement or report. If they later change, then it will be necessary to review the interpretation.

Level III propositions, the top level, relate to an actual offence. These represent the highest level in the hierarchy and they correspond to those which a jury would consider in the event of a trial. Note that these are also level II or "activity level" propositions also. Often, the distinction between the level II and level III propositions can be fine and this is illustrated by comparing IIC and IIIC: the fact that Mr C kicked Mr Z does not necessarily mean that he committed a crime. However, the activity of "assault" is necessarily a crime. Like level II propositions, level III propositions do not depend on evidential material being found.

With level III propositions there are, in general, considerations which are completely outside the domain of the scientist. Of prime importance, of course, is whether or not a crime occurred at all and in some cases that may come down to deciding on the reliability of the word of one witness against another. The scientist is normally in the position of assisting the jury in determining the truth in relation to propositions which are intermediate to the *ultimate issue*, the phrase which is clearly explained by Robertson and Vignaux [8]. This paper considers how the scientist settles for propositions which are as helpful as possible to the jury, without venturing beyond the domain of his/her expertise and knowledge.

Discussion of the hierarchy

Although the classification of propositions into three categories is useful, it does not follow that there are rigid lines of demarcation between them. For example, it has already been mentioned that example IIC is close to being level III pair.

Note that the discussion is confined to cases in which it is meaningful to consider propositions in pairs. There will be

situations where there will be three or more; this happens quite often with DNA mixtures, for example, but it is generally possible to reduce the number of propositions to two which will be identified with the respective prosecution and defence positions. Clearly the two propositions must be mutually exclusive. At first sight, it might be tempting to specify that they are exhaustive and the simplest way to achieve this is by inserting the word "not" into the first proposition, for example:

Mr C is the man who kicked Mr Z

Mr C is not the man who kicked Mr Z

However, this gives the court no idea of the way in which the scientist has assessed the evidence with regard to the second proposition. Mr C might not have kicked the victim, but he might have been present at the incident. However, if he states that he had nothing to do with the incident then this sets the second proposition more precisely to that which we have given at example IIC. Likewise, with example IIB the pair of propositions might simply be stated as:

Mr B had sexual intercourse with Miss Y

Mr B did not have sexual intercourse with Miss Y

However, if semen has been found on the vaginal swab then it may be inferred that *someone* has had sexual intercourse with Miss Y and, indeed, the typing results from the semen would be evaluated by considering their probability given that it came from some other man. It will help the court if this is made plain in the alternative proposition that is specified. It is possible to go further. After consultation with the investigator, the scientist might learn that there appears little prospect of any close relative of the suspect being involved. In the light of this, it seems reasonable to calculate the probability that some person unrelated to Mr B had sexual intercourse with Miss Y and this should later be made clear to the court by expressing the alternatives as:

Mr B had sexual intercourse with Miss Y

Some unknown man, unrelated to Mr B, had sexual intercourse with Miss Y

In summary, the simple use of "not" to frame the alternative proposition is unlikely to be particularly helpful to the court. The prosecution and defence propositions are exhaustive only within the framework of circumstances as they are perceived by the scientist at the time of examination.

It has been stated that the framework of circumstances is made up from the information that is put to the scientist. But, of course, all of that information must later be put to a court of law when it may, or may not, be accepted: if it is accepted then it is evidence, if it is not then it can no longer form part of the framework of circumstances within which the scientist carried out the interpretation. If that occurs then the interpretation must be reappraised.

Expert judgement provides something extra to the observations which a court, unaided, could not do as well and epitomises the value which is added to the process by the scientist. The extent of the added value is determined in part by the closeness of the propositions addressed by the scientist to those addressed by the jury. The less information provided to the scientist in relation to the incident and the suspect, the lower the level of the propositions he can address. Inevitably, the less added value will accrue from his examination. It is also clear that to ascend the hierarchy of propositions requires information. The higher the level of the propositions, the more circumstantial information is required. This means that the interpretation is more vulnerable to change if, later, that information fails to become evidence.

Note the related point that the higher the level of the propositions, the more the expertise of the scientist must come into play. For example, Level II propositions may require expertise in assessing probabilities of transfer and persistence, in addition to that required for carrying out the comparative phase necessary for Level I propositions.

Case example 1

An eyewitness saw a man stand in front of a window and smash it with a case opener. The man ran away and, shortly after, the police apprehended the suspect after a chase down a nearby street. He denied ever having been near to the scene. He is a student with no previous record. His clothing was taken for examination about one hour after the incident. The police provided the scientist with full details of the circumstances, including the size and position of the window.

Let us start with a pair of level II propositions. A feature of the underlying model, outlined in the previous paper, is that these should, as far as possible, be formulated *before* any examination.

The suspect is the man who smashed the window.

The suspect was not present when the window was smashed.

To assist any subsequent court in evaluating these competing propositions, it will be necessary for the scientist to address the probability of whatever evidence has been found given each proposition. To anticipate addressing the relevant questions, assume that the examination has revealed a quantity Q of fragments which match the crime scene. (In the previous paper, there were three categories for Q: *none*, *few* and *many*. But this is not important for the present discussion.)

The first question, leading to the numerator of the likelihood ratio (LR) is:

What is the probability of finding quantity Q of matching glass fragments if the suspect is the man who smashed the window?

In this case, the scientist appears well placed to address this question. In particular, the considerations of glass transfer and persistence can be addressed because the circumstances include details of how the offender smashed the window and how soon after the incident the suspect's clothing was taken.

The second question, leading to the denominator of the *LR* is:

> What is the probability of finding quantity Q of matching glass fragments if the suspect was not present when the window was smashed?

The scientist appears well placed to answer this question also. In particular, the survey of McQuillan and Edgar [6] is relevant to asking questions about the glass to be found on the clothing of young people who are unconnected with any particular crime.

The evaluation of this kind of case is discussed in more detail in Evett and Buckleton [9].

Case example 2

A window was smashed at an unknown time while the owner of the premises was absent. There was no eyewitness. The police attended the scene, toured the surrounding streets and found the suspect, who was known to them, loitering suspiciously in a doorway. He was abusive and refused to give any account of his movements whatsoever. He was known to have an extensive record for criminal damage, breaking and entering. His clothing was taken for examination.

Consider the case at preassessment stage. It is necessary to think of the propositions which might be addressed if a quantity Q of glass is subsequently found which matches the broken window. Consider first the level II propositions that were addressed in the previous case:

> The suspect is the man who smashed the window.
>
> The suspect was not present when the window was smashed.

The first question, as in the previous case is:

> What is the probability of finding quantity Q of matching glass fragments if the suspect is the man who smashed the window?

But this question poses several problems:

1. The question is loaded with an assumption which may be unjustified. Not only is it not known that the window was smashed by a *man* but it is not even known that it was smashed by a *person*! For example, it might have been smashed by a vehicle which had accidentally been reversed into it.

2. There is no information on how the window was smashed. If it was smashed by a man, then did he stand close to it, as in the previous case, or did he stand across the road and throw a brick through it? Without this knowledge, it is difficult to assess the probability of transfer of glass from window to the offender.

3. There is little information about the time of the alleged offence so it is also difficult to assess the probability of persistence of any transferred glass.

The second question is:

> What is the probability of finding quantity Q of matching glass fragments if the suspect was not present when the window was smashed?

There are problems with this question also:

1. The suspect has not claimed that he was not present when the window was smashed, as did the suspect in the first case. It may be that he was present as the window was smashed by some other person.

2. Knowing the background of the suspect, the chance of finding glass for reasons unconnected with this particular incident is not the same as that for a person who appears to have no connection with crime.

At pre-assessment stage, the scientist must also be further influenced by this last point because the nature of the suspect's background means that he/she can expect to find glass on his clothing from other incidents. So, if matching glass is found, its impact can be expected to be diluted by the presence of other non-matching groups of glass fragments.

The scientist, following the principles discussed in the previous paper, would discuss with the customer the expected outcome of an examination and the expected difficulties of interpretation. This case would not appear to offer a particularly good prospect of value for money but imagine that, nevertheless, the customer asks the scientist to proceed.

After examination, a quantity Q of matching glass is found. In this case, however, the scientist is not in a position to assess the evidential weight of Q because of all of the uncertainties associated with the circumstances of the alleged incident. Before starting the case, the scientist has realised that there are questions associated with the interpretation of the presence of glass which cannot be answered. Consequently, it is necessary to settle for level I propositions.

> The glass came from the broken window.
>
> The glass came from some other broken glass object.

These will be addressed solely on the basis of the physical properties of the glass found. All of the issues relevant to addressing the higher level propositions must be left to the court. Let us imagine that the physical properties of the glass are such that the scientist concludes that there is evidence to support the former alternative. If the case subsequently comes to court then prosecuting counsel faces

the task of urging the jury to come to the conclusion that the suspect was responsible for smashing the window, in the light of the scientific evidence and the non-scientific evidence. The scientist might be asked to contribute to the argument by answering questions about hypothetical situations such as, for example, "is this the quantity of glass you would have expected if Mr X had smashed the window ten-minutes before he was picked up?". Defence, likewise might suggest other hypothetical situations for the scientist to consider such as "what would be your expectation of finding glass on Mr X's clothing if you were told that he had been working on a demolition site that day?". There is a blurring of the roles of scientist and advocate here.

Case example 3

This example is taken from a recent paper on footwear evidence by Evett, Lambert and Buckleton [10].

A burglary was committed in Leicester, entry being gained via a forced window from a flat roof. A left footwear mark was found at the point of entry. There was no immediate suspect, but the mark was encoded and stored on computer. A month later, a man was arrested on suspicion of a burglary in Market Harborough. He was found to be wearing a pair of size nine Adidas Galaxy shoes. Prints of the soles were searched against the database of marks from unsolved crimes and that from the Leicester burglary was retrieved. The suspect's home was searched and a few items of property were found there which were similar to items taken during the burglary.

The shoes and a lift of the scene mark were submitted for expert examination. The mark was found to agree in pattern with the left shoe and, although incomplete, it corresponded in size. There were a few small cut marks, which were also present on the suspect's left shoe. However, the suspect's shoe was rather more worn and also bore a few marks which were not present in the scene mark.

Records kept at the laboratory showed that approximately 1% of footwear that had previously been submitted for examination in crime cases would be considered to be indistinguishable in pattern and size from the scene mark in this case.

Evett et al [10] started by considering the two level III propositions:

C: the suspect is the offender

$\bar{C}$: some unknown person is the offender

A formal analysis led to the expression:

$$LR = rw.lr_U.lr_M$$

Where the terms may be briefly explained as follows:

r: This is the probability that the footwear mark was left by the offender.

w: The probability that the suspect would have been wearing that particular pair of shoes if he had been the offender.

lr_U: This is a likelihood ratio which takes account of the evidential weight of any "acquired features" in the scene mark. These are features which do not derive from the manufacture of the shoe, but from wear and accidental cuts. The likelihood ratio is calculated from the answers to the two questions:

> What is the probability of the observed acquired features in the scene mark if it had been left by suspect's left Adidas Galaxy?
>
> What is the probability of the observed acquired features in the scene mark if it had been left by some other left Adidas Galaxy?

lr_M: This also is a likelihood ratio and this takes account of the evidential weight of the manufacturing features of the scene mark. It is the ratio of the answers to these two questions:

> What is the probability of the observed manufacturing features in the scene mark if it had been left by the suspect's left shoe?
>
> What is the probability of the observed manufacturing features in the scene mark if it had been left by some other left shoe?

There is no doubt that a court would see lr_U and lr_M as legitimate areas for the scientist to express an opinion. But what about r and w? In this particular example, the fact that the mark was found at the point of entry might enable the scientist to give the opinion that it was almost certainly left by the offender, so $r \simeq 1$. On the other hand, w is more difficult – assessing the chance that the suspect would have been wearing that particular pair of shoes if he had committed the offence involves issues that are by no means clear. The solution to this might be that it should be left to the jury. Effectively, this means that the scientist is assisting the jury by addressing the level I propositions inherent in the evaluation of lr_U and lr_M:

A: the scene mark was made by the suspect's left shoe

$\bar{A}$: the scene mark was made by some unknown shoe

Then, to get to the jury level, it is necessary for the court to address two questions:

> What is the probability that the mark was left by the offender (r)? (In this case, the scientist may be seen to be the correct source of advice)
>
> If the suspect were the offender, then what is the probability that he would be wearing that pair of shoes at the time (w)?

Case example 4

Two masked men armed with sawn-off shotguns burst into a sub-post office and threatened the counter staff. The takings for the day were handed over and the men left.

Witnesses inside the post office said that one man was wearing a dark green balaclava type mask and the other was wearing a knotted piece of stocking/tights over his head and face. Witnesses outside the post office said that the two men were driven away from the scene at high speed in a car driven by a third man. The witnesses saw various articles being discarded from the car as it drove off and a later search by police officers led to the recovery of some items of clothing. Some way further along the supposed getaway route, a dark green balaclava was found. There is no witness to say that this item was discarded from the vehicle.

Following a tip-off from an informant, Mr U was arrested three weeks after the offence. He denied all knowledge of the incident. A sample of hair was taken and submitted with the balaclava mask for examination.

Once again, consider the position of the scientist before he carries out an examination. A possible pair of level III propositions is:

The suspect is one of the offenders

The suspect is unconnected with the offence

This pair will be very close to the deliberations of a jury, if the case comes to court. Let Q denote the quantity of hairs found from an examination of the balaclava which match the hair of Mr U, where Q could take a range of values, perhaps extending between "zero" and "many". Then the question to be addressed with regard to the numerator of the *LR* will be "what is the probability of finding Q matching hairs if Mr U is one of the offenders?" This carries the implicit acceptance that an offence has taken place and that one of the offenders wore a balaclava. There is a problem, however, because there is uncertainty about whether or not the balaclava that was found is the one that was worn by one of the offenders. It is difficult to know how this uncertainty is to be addressed but one thing is clear – it is outside the scientist's expertise. So let us instead frame a second pair of propositions:

The suspect has worn the balaclava

The suspect has never worn the balaclava

These are also level II propositions, though they are further from the jury level propositions than were the first pair. If Q matching hairs are subsequently found, then the numerator of the *LR* will be addressed by the question "what is the probability of finding Q matching hairs if Mr U has worn the balaclava?". Once again it is necessary to think carefully about whether the scientist will be in a position to answer this question. What is known about the transfer of hairs under such circumstances? How long was the balaclava worn for? It may be that the uncertainties are such that the scientist must settle for the lowest level of proposition. If Q is not zero, the pair of propositions are:

The hairs are those of the suspect

The hairs are from another person

The *LR* for this pair will be a function of the closeness of the match and of the scientist's judgement of the rareness/commonness of the observed hair type. It is worth emphasising here that lack of data does not prevent the scientist from addressing the questions. If there are data, all well and good, but if there are not then it is reasonable to rely on the expert judgement of the scientist. Note that if this case subsequently comes to court, all of the inferential processes between these two low level propositions and those addressed by the jury will be left to the court. No doubt, prosecution and defence will attempt to sway the jury with speculative arguments relating to the mechanisms by which hairs may have got from Mr U to the balaclava; and relating to the finding of the balaclava and its possible association with the crime. The scientist cannot be of direct help here; though, as we discussed in case example 2, he may assist one or both of the counsel by advising on his expectations given hypothetical situations which may be put to the jury.

If, on the other hand, no matching hairs are found, the scientist will have to say that he is unable to interpret their absence. Again it will be a matter for advocacy, if the case ever comes to court.

Case example 5

A witness saw two men who were fighting each other in a street, late at night. As he watched, one of the men fell to the ground and the second man then proceeded to kick him several times about the upper body and head. The man on the ground was unconscious and bleeding from head wounds. The other man made off. It later transpired that the injured man, Mr Z was one of two who had been seen by witnesses to have been arguing violently in a nearby pub. On this information, the other of the two, Mr C, was arrested at an all-night cafe shortly after the incident. He said that he and Mr Z had parted amicably outside the pub and he had no knowledge of the fact that Mr Z had been injured in a fight.

Mr C's clothing and shoes were taken for examination, together with blood samples from Mr Z and Mr C.

Consider this first at pre-assessment stage. A potential prosecution proposition is:

Mr C is the man who kicked Mr Z

On the other hand, Mr C denies all knowledge of the fight and so the defence proposition is:

Mr C was not present at the kicking of Mr Z

These are level II propositions, as at example IIC, and the scientist should consider his/her expectations given each of them. If the prosecution proposition is correct then Mr Z bled profusely and blood would be expected, probably in

the form of spatters, on Mr C's shoes and trousers. If the defence proposition is correct then the survey of Briggs [5] shows that it is very unlikely that blood other than Mr C's own would be found on his clothing. So, blood spattering to match Mr Z and consequent large *LR*s are expected if the prosecution proposition is correct; and no matching blood and *LR*s less than one are expected if the defence proposition is correct.

If the examination subsequently proceeds and blood spattering is indeed found on Mr C's shoes and trousers which matches Mr Z, then there are two questions:

> What is the probability of finding matching blood spattering of the observed extent if Mr C is the man who kicked Mr Z?
>
> What is the probability of finding matching blood spattering of the observed extent if Mr C was not present at the kicking of Mr Z?

Note that there are two strands of evidence to evaluate: the typing of the blood; and its distribution. Evett and Buckleton [11] have discussed the evaluation of such a case in more detail.

Case example 6

A pedestrian found a man lying unconscious in the street. The man appeared to have suffered head injuries and he was bleeding profusely. The investigator found that the man, Mr Z, had recently been seen leaving a nearby pub with another man, Mr C, with whom he was arguing violently. Mr C was arrested shortly after the incident and he refused to respond to any questions in relation to the incident, nor in relation to his own movements.

This case differs from example 5 in two respects. First, there is no eyewitness to the incident that led to Mr Z's injuries, thus it would be ingenuous to start with a prosecution proposition of the form:

> Mr C is the man who kicked Mr Z

This is because it is not even known that Mr Z was kicked at all: he might, for example, have been injured by a passing vehicle. Second, in this case Mr C gives no account of his movements and does not suggest any particular line of defence and so the defence alternative:

> Mr C was not present at the kicking of Mr Z

is also ingenuous, not only because it is not known that there was a kicking, but also because Mr C might well have been present at the incident as a bystander without actively participating in the kicking.

The lack of circumstantial information in this case means that the scientist should advise the investigator that it is necessary to address propositions at a lower level than those considered in Case 5: propositions of the kind IC in the table, for example. But, of course, knowing so little of the incident, it is not possible to advise the investigator of how much blood, if any, can be expected to be found.

Imagine that, in spite of this uncertainty, the investigator asks the scientist to proceed with an examination of Mr C's shoes and trousers to see if there is blood present on them. Examination reveals poorly defined areas of bloodstaining which is found to be of the same type as that of Mr Z. Now, in view of the uncertainties we face about the circumstances of the incident and the absence of any explanation from Mr C, it would seem prudent to address level propositions of the kind:

> The blood on Mr C's clothing and shoes came from Mr Z
>
> The blood on Mr C's clothing and shoes came from some unknown person

The *LR* for this pair will be determined solely by the match probability calculated from the genotype of the staining.

There is an interesting development of this example. Imagine that the staining consisted of a pattern of spots which had an appearance similar to that which had been observed in experiments to simulate kicking bleeding bodies. This observation is clearly evidence in itself and may influence the alternatives that the scientist addresses. However, this is not a straightforward matter to deal with: it is similar to the issues that arise in reconstruction problems, a subject which will be left to a future paper.

Resumé

The important points made in this paper can be summarised as follows:

1. To express an opinion about the evidential weight associated with the presence or absence of transferred material, the forensic scientist must consider a pair of propositions which will represent, respectively, the prosecution and defence positions.

2. Pairs of propositions can be seen to fall into three broad categories in a three-level hierarchy with the following generic designations: Source; Activity; and Offence.

3. To address higher level propositions, the scientist needs to be given information which leads to a *framework of circumstances*. There may be need for direct interaction between scientist, investigator and defence team in order to formulate the pair which will be of greatest assistance to the court.

4. The framework of circumstances is provisional up to the point where the court decides what information is accepted as evidence and until the defence position is established. If the framework changes then the interpretation must be reviewed.

5. The higher the level of propositions that are addressed, the greater the reliance on the expertise of the scientist.

6. The higher the level of propositions that are addressed, the greater the value added by the scientific evidence.

7. If the information available to the scientist is such that it is necessary to address level I propositions, then the inferential steps needed to address the level III, or jury level, propositions will be influenced by the advocacy of prosecution and defence counsel.

The next paper in this series will discuss further aspects of the case assessment and interpretation model which was introduced in the previous paper. The ideas are continuing to evolve and expand as they are explored in case-based workshops.

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